

Discussion 2 Posts

Initial Post

DNS Security Extensions (DNSSEC)

DNSSEC are a set of protocols that work to confirm the zone of origin and maintain integrity of data received from DNS queries. This is done by adding digital signatures from each domain zone of DNS' hierarchical structure to the data, creating a chain of trust (ICANN, 2019). Adding this confirmation of authenticity prevents attacks such as DNS spoofing, or cache poisoning, where attackers would alter records to direct users to malicious websites (Shelley & Gibson, 2023). DNSSEC is a primary method of mitigating the Kaminsky attack, a type of cache poisoning discovered in 2008 (Alexiou et al., 2010).

Despite DNSSEC being a useful tool for combating problems relating to the integrity of data, it doesn't provide protections relating to data confidentiality (Schmid, 2021). There are also concerns that it can be used to amplify DoS attacks, lowering the availability of data (van Rijswijk-Deij, 2014).

These concerns, and the complicated nature of configuring DNSSEC, has led to some hesitancy in adoption of DNSSEC (Chung et al., 2017), with around 35% of DNS queries being validated by DNSSEC worldwide (Apnic Labs, 2024).

Overall, despite these negatives, DNSSEC is a valuable tool and worth implementing; it addresses an inherent weakness in DNS and protects against damaging attacks that would otherwise have no mitigation.

Honeypots

Honeypots are systems set up to trick potential attackers into thinking they have gained access to sensitive data, with the goal of monitoring the attacker's actions. Honeypots vary in size and scope, they can be solitary systems configured to look and behave like other systems, but they can also be whole networks; these are more often called honeynets (Joshi & Sardana, 2011).

Honeypots are a way of gathering data on potential attackers and their exploits by tracking their movements and analysing their techniques whilst in a system. They also have the benefit of distracting malicious actors from servers and networks containing real or useful data (Shelley and Gibson, 2023).

Honeypots have low network activity when compared to other systems, like IDS, which monitor all traffic. The purpose of a honeypot is to collect data on malicious activity, therefore all activity on these systems are the work of an attacker (Provos & Thorston,

2021). Consequently, honeypots require low resources and should create a low rate of false positives.

A negative aspect of honeypots are that they only monitor activity inside the fake system or network. If attacks are happening elsewhere honeypots will not know. An additional concern are situations where attackers become aware of the honeypot's existence. They can then feed false information into that system, obscuring their true methods and intentions (Spitzner, 2002).

When used correctly honeypots are a useful technology to analyse the methods and motivations of attackers and add a proactive element into system security. Problems can arise when honeypots are relied upon to do more than their primary purpose, or are not implemented properly, leading to detection. Accordingly, honeypots should be used as part of a broad defensive strategy, alongside other more reactive technologies, like firewalls and IDS, that analyse all traffic.

References:

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Peer Reviews

Peer Review 1:

The initial post introduced Intrusion Detection System (IDS) and Public Key Infrastructure (PKI) technologies.

The post details that we can divide IDS into two categories: Network-based IDS (NIDS), for individual systems and devices, and Host-based IDS (HIDS), that operate over the network. However, these definitions should be switched, as it is NIDS that works over the network and HIDS that works for individual systems and devices (Efe & Abaci, 2022: 24).

It is also possible to divide IDS types by detection method. Signature / misuse-based detection compares traffic to a list of known attack signatures, and anomaly-based detection sets what it considers to be normal behaviour for its environment and checks for anomalous behaviour. Aspects of both can be combined for a hybrid detection type of IDS (Abbas, Naser & Abbas, 2023: 156).

To further analyse the topic of IDS types we could ask which category of IDS is right for different users, and what are the strengths and weaknesses of the different detection methods?

The discussion on PKI technology covers its nature and methods. It makes good points about how it is used to maintain aspects of the CIA pyramid and covers some drawbacks of the technology.

To go deeper into PKI and its application, use-cases could be analysed. For example, SSL and TLS use PKI technology to help secure web traffic and authenticate servers (van Oorschot, 2021:229).

Word Count: 230

References:

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Peer Review 2

The initial post has provided an interesting and well-researched overview of firewall technology and has introduced some ideas on the most common types of firewall and considerations affecting their adoption.

For greater clarity it would be useful to introduce more modern technologies such as proxy and Next-Gen firewalls, alongside some explanation of the differences between these various types. For example, both packet filtering and stateful firewalls perform packet filtering; packet filtering firewalls analyse packet headers and compare them to rules in an ACL before deciding on how to proceed, whereas stateful firewalls consider the context of traffic and network connections first, before proceeding to an ACL if needed (Johnson, 2016). How do proxy firewalls and next-gen firewalls make decisions on filtering traffic?

The initial post's point about packet filtering firewalls being mainly used by small businesses would also benefit from further analysis, by linking the adoption considerations mentioned in the post to technology specific strengths and weaknesses. In this case it should be mentioned that packet filtering firewalls are thought of as the least secure type of firewall (Madhloom et al., 2023). Why then are they the most popular for small businesses? Further analysis on which firewall type is most suitable for individuals, large organisations and different real-world situations would also enhance the post.

Data loss prevention (DLP) systems were also mentioned. Whilst some challenges facing DLP were included, discussing the challenges of implementing DLP systems would add depth. These include balancing security and productivity, a high degree of false positives and overall system complexity (Machado de Sousa & Shahzad, 2021).

Word Count: 261

References:

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Madhloom, J.K., Noori, Z.H., Ebis, S.K., Hassen, O.A. and Darwish, S.M. (2023). An Information Security Engineering Framework for Modeling Packet Filtering Firewall Using Neutrosophic Petri Nets. *Computers* 12(10): 202. DOI: <https://doi.org/10.3390/computers12100202>.

Peer Review 3:

The initial post has provided a detailed introduction to Intrusion Detection System (IDS) technologies, discussing its history and development. It also mentions an interesting use-case in which IDS can be used to help companies meet regulatory requirements.

The post details how IDSs can be classified by data source, or by detection method, with the strengths and weaknesses of the options being considered. The topic could be expanded by adding another detection method gaining popularity - hybrid signature / anomaly-based IDS (Alharbi & Khan. 2023) and analysing how it mitigates some weaknesses of both individual approaches.

An interesting point is made regarding IDS not being enough on its own to prevent security threats and needing to be integrated with Intrusion Prevention Systems (IPS). IPSs take a proactive approach to security problems, attempting to stop attacks, but also contain the capabilities of an IDS (Erney & Chowdhury, 2022) so could be considered an IDS replacement. For redundancy purposes, however, they could be used together as part of a broader defence in depth strategy.

To go deeper into the topic, the functionality of IDS / IPS can be evaluated against other defensive technologies, such as firewalls and Security Information and Event Management (SIEM) solutions. The value of implementing these alongside IDS / IPS can also be debated.

Word Count: 214

References:

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Summary Post

The initial post introduced DNSSEC and honeypots, two security technologies.

DNSSEC was described as a way of protecting DNS data integrity and the main protection against DNS spoofing. Concerns included DNSSEC use in DOS attacks and configuration difficulties, leading to a low adoption rate. The initial post's conclusion was that DNSSEC is worth implementing, as it provided protection that had no other strong mitigation.

Huston (2024) disagrees, believing that without changes DNSSEC should be abandoned, with the low adoption rate cited as a reason. This is a valid point; however, it is based heavily on the validation rates in the US, UK and Australia, and ignores the fact that in some countries, such as New Zealand, Sweden and Germany, the DNSSEC validation rate is 80% or higher. Overall Europe rose 8% year-on-year, another positive sign (Apnic Labs, 2024).

Given that the DNSSEC usage is climbing, albeit slowly, and its rise in Germany and Europe generally, the initial post's conclusion that DNSSEC is worth implementing should be maintained.

Honeypots were described as a proactive security technology that allow malicious actors access to a system containing useless information to monitor attack methods. Positives were low resource consumption and low rate of false positives; negatives are a narrow field of focus and risk of attackers discovering the honeypot and providing misleading information. This led to the recommendation that honeypots should be used alongside other security technology.

Neary (2024) suggested locating a honeypot inside a DMZ, which is logical. This placement should ensure attackers cannot use their honeypot activity to enter internal systems and access protected information. This approach, combined with firewalls on both ends of the DMZ, was tested, and found to bring greater security (Nikoi, et al., 2022: 25). The success of this test emphasises the original conclusion that honeypots are best used as part of a defence-in-depth strategy.

Word Count: 307

References:

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